

Daniel Gordon

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<https://danielgordon10.github.io/>

Education

The University of Washington

2014 – 2020

Ph. D. in Computer Science

Advised by Dieter Fox and Ali Farhadi

The University of Washington

2016

Masters in Computer Science

Washington University in St. Louis

2010 – 2014

Bachelor of Science in Computer Science

Second Major in Entrepreneurship

Summa Cum Laude – GPA: 3.96, Engineering Class Rank: 8/323

Undergraduate Research Advised by Robert Pless

Work Experience

Apple Maps 3DV:

Apr 2024 – Present

Senior Machine Learning Engineer

- Responsible for diffusion, semantic segmentation, and detection models being run on imagery at world scale
- Developed end-to-end models for predicting 3D vectorized maps from raw pixels

Third Wave Automation:

Team Lead: Perception and Pallet Manipulation

Jun 2023 – Mar 2024

- Oversaw a team of 5 engineers serving as both a manager and a contributor for several major projects
- Facilitated cross-team collaboration resulting in doubling the success rate of autonomous workflows

Tech Lead: Perception

Jan 2022 – Dec 2022

- Led a team of 3 engineers working on major features for the core competencies of the product
- Contributed technical design expertise throughout the code stack
- Pioneered Python type checking throughout the codebase

Independent Contributor: Perception

Jun 2020 – Dec 2021

- Designed and implemented many visual algorithms for the core product including object detection and tracking, pose estimation, local mapping and collision checking, and several safety systems

Facebook AI Research (FAIR):	Winter 2019
Research Intern – A-STAR Team with Dhruv Batra	
Conducted research resulting in the ICCV publication “SplitNet: Sim2Sim and Task2Task Transfer for Embodied Visual Navigation”	
Allen Institute for Artificial Intelligence:	Winter 2017
Research Intern – PRIOR Team with Roozbeh Mottaghi	
Conducted research resulting in the ICCV publication “Visual Semantic Planning using Deep Successor Representations”	
Google:	
Software Engineering Intern – Google Maps	Summer 2013,
- Designed and programmed the Street View Time Machine frontend - Increased polish and feature improvement on the new Maps frontend	Summer 2014
Engineering Practicum Intern – Google Wallet	Summer 2012
- Integrated an autocomplete feature to the Wallet website - Added Google+ profile images and names to various Wallet pages - Created the Wallet dashboard page and recent transaction widget	

Publications

Learning by Watching and Learning by Doing	2020
Ph.D. Thesis	
Watching the World Go By: Representation Learning from Unlabeled Videos	Arxiv 2020
Daniel Gordon, Kiana Ehsani, Dieter Fox, Ali Farhadi	
What Can You Learn from Your Muscles? Learning Visual Representations from Human Interactions	Arxiv 2020
Kiana Ehsani, Daniel Gordon, Thomas Nguyen, Roozbeh Mottaghi, Ali Farhadi	
ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks	CVPR 2020
Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, Dieter Fox	
SplitNet: Sim2Sim and Task2Task Transfer for Embodied Visual Navigation	ICCV 2019
Daniel Gordon, Abhishek Kadian, Devi Parikh, Judy Hoffman, Dhruv Batra	
What Should I Do Now? Marrying Reinforcement Learning and Symbolic Planning	Arxiv 2018
Daniel Gordon, Dieter Fox, Ali Farhadi	
Shifting the Baseline: Single Modality Performance on Visual Navigation & QA	NAACL 2019
Jesse Thomason, Daniel Gordon, Yonatan Bisk	Short

IQA: Visual Question Answering in Interactive Environments Daniel Gordon, Aniruddha Kembhavi, Mohammad Rastegari, Joseph Redmon, Dieter Fox, Ali Farhadi Received the Nvidia Pioneering Research Award at CVPR 2018	CVPR 2018
AI2-THOR: An Interactive 3D Environment for Visual AI Eric Kolve, Roozbeh Mottaghi, Daniel Gordon, Winson Han, Eli VanderBilt, Luca Weihs, Alvaro Herrasti, Matt Deitke, Kiana Ehsani, Yuke Zhu, Abhinav Gupta, Ali Farhadi	Technical Report 2017
Re3: Real-Time Recurrent Regression Networks for Object Tracking Daniel Gordon, Ali Farhadi, Dieter Fox	RA-L 2018
Visual Semantic Planning using Deep Successor Representations Daniel Gordon, Yuke Zhu, Eric Kolve, Dieter Fox, Li Fei-Fei, Abhinav Gupta, Roozbeh Mottaghi, Ali Farhadi	ICCV 2017
Collaborative Rephotography Ruth West, Abby Halley, Daniel Gordon, Jarlath O'Neil-Dunne, Robert Pless	SIGGRAPH 2013 Studio Talks
Collaborative Imaging of Urban Forest Dynamics: Augmenting Rephotography to Visualize Changes over Time Ruth West, Abby Halley, Jarlath O Neil-Dunne, Daniel Gordon, Robert Pless	IS&T/SPIE 2013

Service

Co-organizer of 1st Workshop on Visual Understanding Across Modalities and THOR competition http://vuchallenge.org/	CVPR 2017
Organizer of Deep Learning in Practice Seminar Talk Series https://sites.google.com/cs.washington.edu/deeplearninginpractice/	Summer 2017

Teaching Experience

Teaching Assistant at the University of Washington Introduction to Deep Learning: Head TA - Wrote Numpy-based Pytorch-like library for deep learning assignments and autograder for grading - Managed 5 other TAs and 160 students	Fall 2018, Fall 2019
Teaching Assistant at Washington University in St. Louis Introduction to Artificial Intelligence Algorithms and Data Structures Logic and Discrete Mathematics Introduction to Computer Science	Spring 2013, Spring 2014 Fall 2013 Fall 2012 Fall 2010-Spring 2012

Honors and Awards

NVIDIA Graduate Fellowship (1 of 10 awardees from 230+ applicants)	2019
National Science Foundation GRFP Honorable Mention (Top 1/3 rd of applicants)	2015 and 2016
Wissner-Slivka Fellowship (University of Washington CSE)	2014
Achievement Rewards for College Scientists Fellowship (UW CSE 1 of 2 awardees)	2014-2016
Outstanding Senior Award – Computer Science (Washington University)	2014
Sigma Xi (Washington University)	Inducted Spring 2014
Upsilon Pi Epsilon (Washington University Top 1/3 rd of CSE Class)	Inducted Fall 2013
Tau Beta Pi (Washington University Top 1/8 th of Engineering Class)	Inducted Fall 2012

Patents

Providing a thumbnail image that follows a main image US Patent 9,934,222	April 3, 2018
Display screen with graphical user interface or portion thereof US Patent D780,795	March 14, 2017

Technical Skills

Proficient in	Python, Java, PyTorch, C++, C, TensorFlow, Caffe, Git, HTML, CSS
Capable in	Bazel, Matlab, Javascript, Google Closure, Android, PHP, Mercurial, C#, LaTeX
Basic Knowledge	CUDA, Objective-C/Cocoa, iPhone, MySQL, JQuery, Unix Terminal

Open Source Repositories

VINCE: <https://github.com/danielgordon10/vince>
YouTube8M Data: <https://github.com/danielgordon10/youtube8m-data>
Deep Learning Class Numpy Library: <https://gitlab.com/danielgordon10/dl-class-2019a>
SplitNet: <https://github.com/facebookresearch/splitnet>
AI-Habitat: <https://github.com/facebookresearch/habitat-api>
AI2-THOR: <https://github.com/allenai/ai2thor>
Re3: <https://github.com/danielgordon10/re3-tensorflow>
Re3-Pytorch: <https://github.com/danielgordon10/re3-pytorch>
IQA: <https://github.com/danielgordon10/thor-iqa-cvpr-2018>
